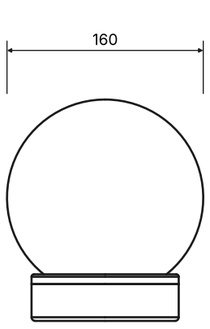
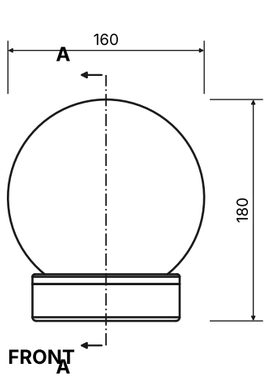
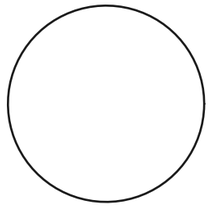
Scale 1:5. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

General arrangement

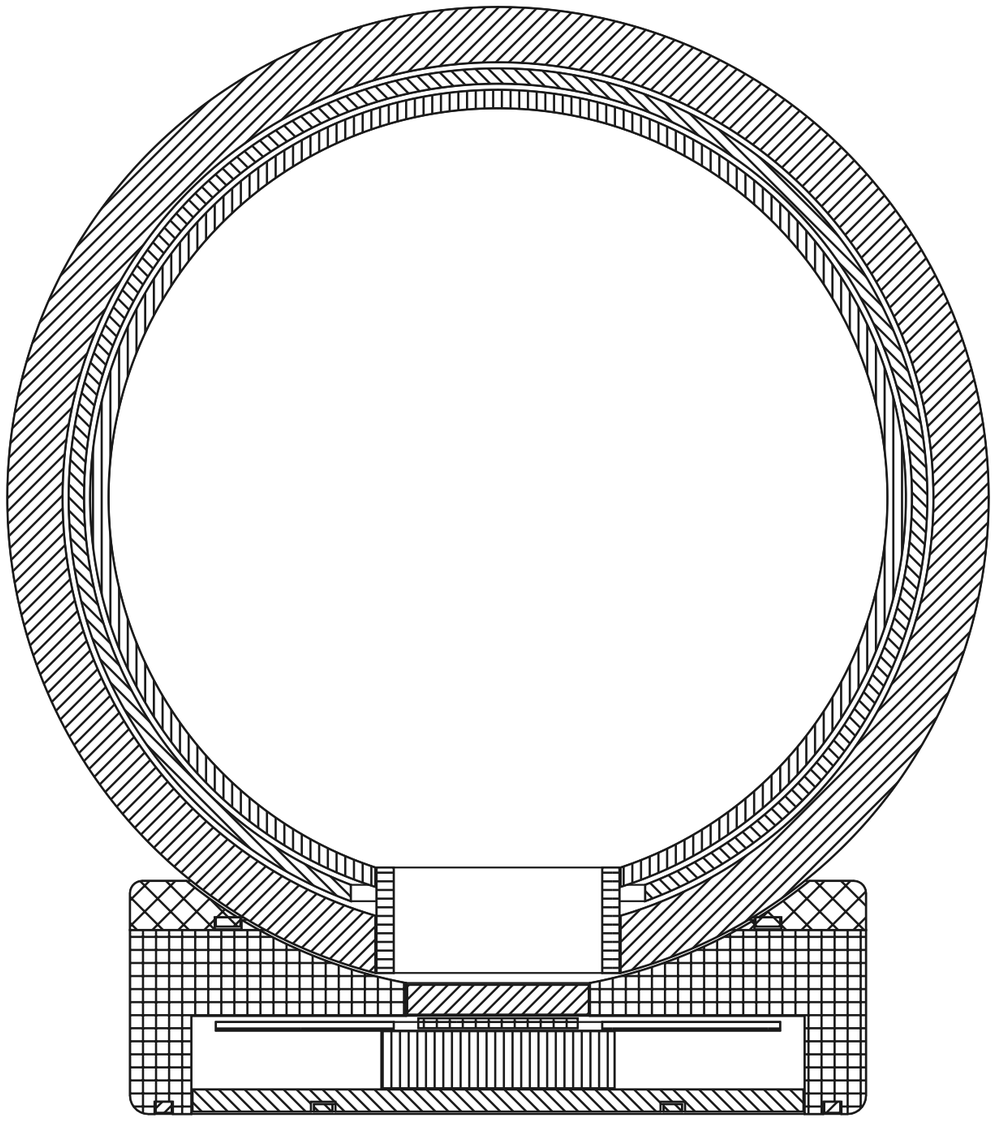
Orb table lamp: Cordless rechargeable table lamp in cast acrylic for bars, a crystal ball

mm | issue 2026-10-19 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

EOT-001 rev A

Scale 1:1. Section A-A (sheet 1): the product cut at x 0.0 and seen from the right; cut faces hatched, a different angle for each part.



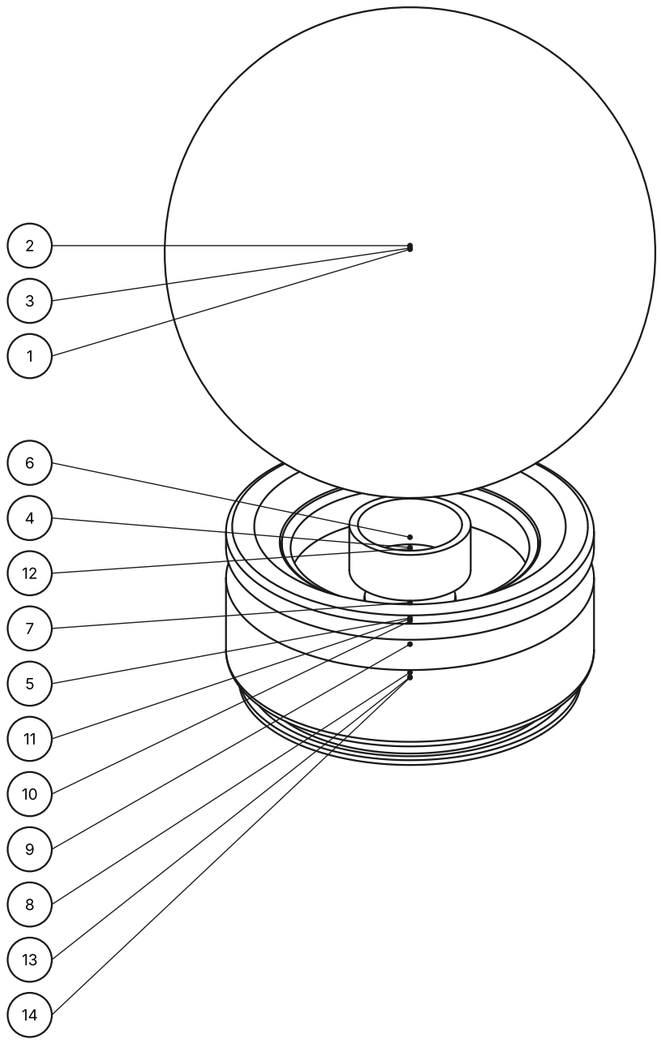
Section A-A

Orb table lamp: Cordless rechargeable table lamp in cast acrylic for bars, a crystal ball

mm | issue 2026-10-19 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2, parts drawn apart. Balloons are the parts list's numbers (last sheet).



Exploded view

Orb table lamp: Cordless rechargeable table lamp in cast acrylic for bars, a crystal ball

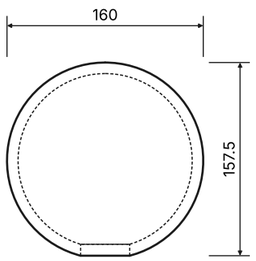
mm | issue 2026-10-19 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

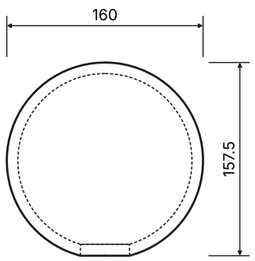
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

1. shell: lathe, cast-acrylic; 634 cm3 each; 1 for the set; 1:5

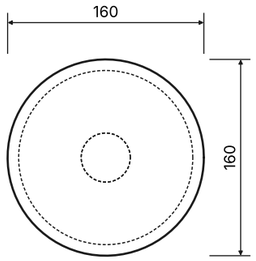
the ball's outside: two hemispheres turned from cast block in the tint, 9 wall, bonded at the equator and polished (or rotationally cast: ask the caster); a 40 port at its underside



FRONT



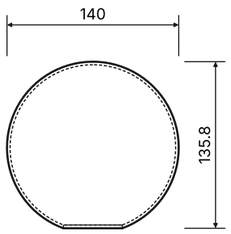
FROM THE RIGHT



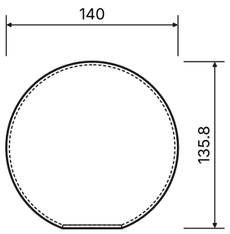
FROM ABOVE

2. liner: lathe, cast-acrylic; 144 cm3 each; 1 for the set; 1:5

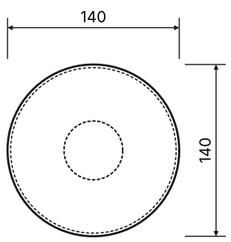
the diffuser: a 2.5 ball in clear, bead-blasted, in two halves inside the shell



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Orb table lamp: Cordless rechargeable table lamp in cast acrylic for bars, a crystal ball

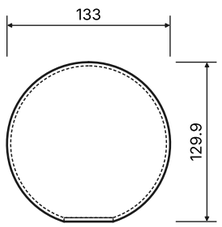
mm | issue 2026-10-19 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

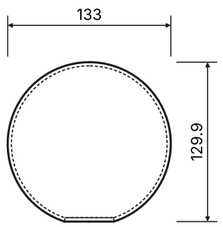
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

3. skin: lathe, cast-acrylic; 155 cm3 each; 1 for the set; 1:5

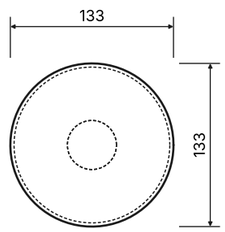
a 3 ball in the tint inside the liner: seen through the port, the frost is still behind acrylic



FRONT



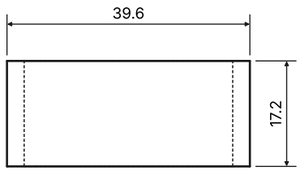
FROM THE RIGHT



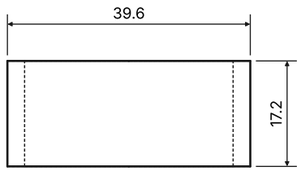
FROM ABOVE

4. grommet: lathe, brass; 6 cm3 each; 1 for the set; 1:1

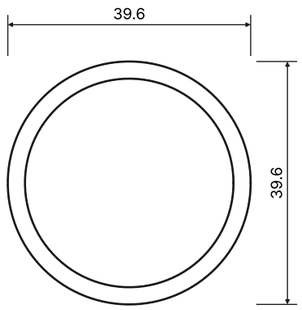
lines the port and closes the three layers there, bonded



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Orb table lamp: Cordless rechargeable table lamp in cast acrylic for bars, a crystal ball

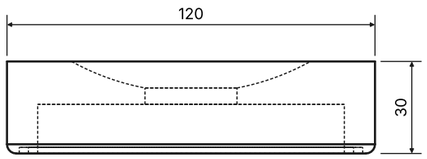
mm | issue 2026-10-19 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

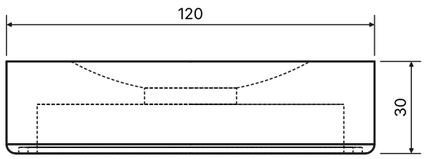
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

5. base: lathe, steel; 183 cm3 each; 1 for the set; 1:2

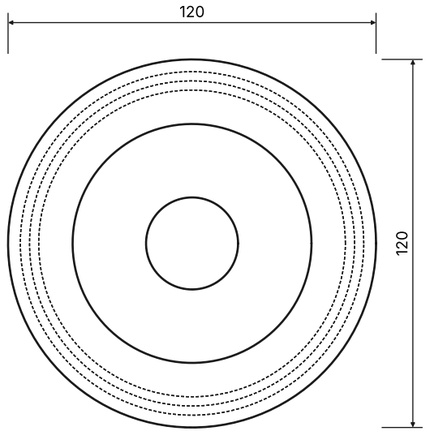
mild steel, blackened and waxed (or black e-coat): 120 × 30, its weight with the ball's about 2.6 kg; the cup turned in it and the ring together



FRONT



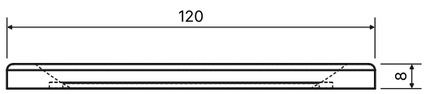
FROM THE RIGHT



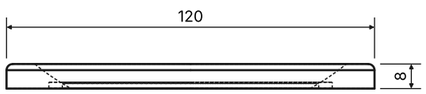
FROM ABOVE

6. ring: lathe, brass; 36 cm3 each; 1 for the set; 1:2

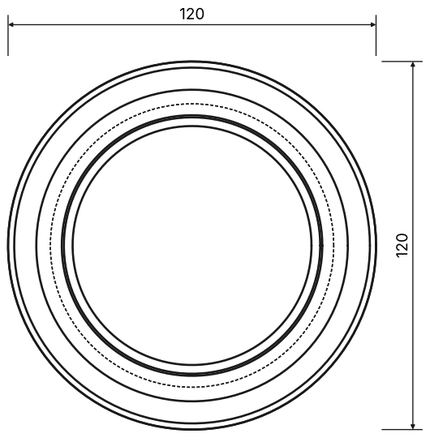
brass, satin, insulated from the base: the touch (tap to switch, hold to dim, 2200 to 2700 K)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Orb table lamp: Cordless rechargeable table lamp in cast acrylic for bars, a crystal ball

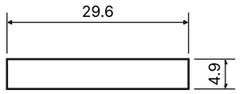
mm | issue 2026-10-19 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

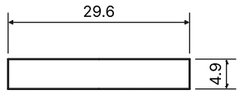
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

7. window: lathe, cast-acrylic; 3 cm3 each; 1 for the set; 1:1

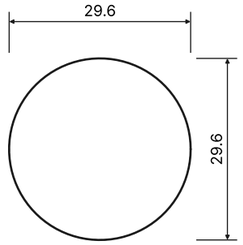
a clear acrylic plug, sealed with an O-ring: the LED under it, the base IP44 with the ball off



FRONT



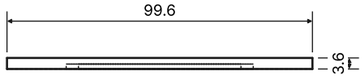
FROM THE RIGHT



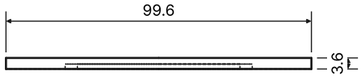
FROM ABOVE

8. plate: lathe, steel; 27 cm3 each; 1 for the set; 1:2

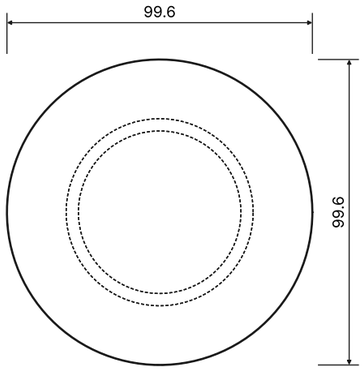
closes the pocket from below on an O-ring; the gold contact rings on it



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Orb table lamp: Cordless rechargeable table lamp in cast acrylic for bars, a crystal ball

mm | issue 2026-10-19 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	shell	lathe	1	Cast acrylic tube, tinted through	170 × 170 × 167	358.58
2	liner	lathe	1	Cast acrylic tube, tinted through	150 × 150 × 146	256.18
3	skin	lathe	1	Cast acrylic tube, tinted through	143 × 143 × 140	239.45
4	grommet	lathe	1	Brass C360	50 × 50 × 27	53.03
5	base	lathe	1	Mild steel	130 × 130 × 40	51.76
6	ring	lathe	1	Brass C360	130 × 130 × 18	81.50
7	window	lathe	1	Cast acrylic tube, tinted through	40 × 40 × 15	45.93
8	plate	lathe	1	Mild steel	110 × 110 × 14	46.63
9	battery	buy	1	Samsung SDI INR18650-35E Li-ion cells, 3,500 mAh, two in parallel in a welded pack with its own protection board and an NTC (7.0 Ah at 3.6 V: about 25 Wh, 12 h at the 2 W bar setting)		14.00
10	board	buy	1	to make (a contract PCB house) The table lamp's board: a USB-C and pogo-pin charger (TI BQ25895 kind), a capacitive touch dimmer on the brass finial (stepless, 2200 to 2700 K warm-dim mixing two LED channels), a low-battery blink,		22.00
11	led	buy	1	Nichia or Seoul (to choose) A 30 mm round warm-dim LED board, two channels (2200 K and 2700 K, CRI 90+), about 2.5 W at full, 200 lm; a second 4-LED ring at the column's foot to light the column (0.3 W)		9.00
12	seat	buy	1	A silicone seat ring 92 × 84 × 1.5 (Shore 40, black) in the base's cup: the ball rests on it, and it stops a nudge rolling it		1.50
13	foot-ring	buy	1	Two EPDM O-rings (the column in the foot, the hub's lid), a silicone foot ring 110 x 3 (Shore 40, non-slip, wipeable), a USB-C port with a rubber plug: IP44		4.00
14	contacts	buy	1	Mill-Max (or alike) Two gold-plated spring-loaded contact pins in the tray, two flat gold rings in the foot's underside, and a 5 V tray of six (the staff's charging tray, its own product)		6.00

Notes

1. A crystal ball: a 160 tinted ball, low in the cup of a 120 × 38 base. The ball lifts off: the LED is under a sealed window in the base, so balls swap colour by hand, as the Dome's covers do.
2. Cordless: two 18650 cells (25 Wh) sealed in the base, about 12 hours at the bar setting; it charges on the staff tray through two gold rings underneath, or by USB-C. Touch the brass ring: tap to switch, hold to dim, 2200 to 2700 K as it dims.
3. IP44: the base sealed by the window's and the plate's O-rings; with the ball on or off. Wipe with a damp cloth.
4. The ball's wall is tint, frost, tint, as the Dome's: the port underneath is lined in brass, and through it the frost is still behind acrylic.

Parts list and notes

Orb table lamp: Cordless rechargeable table lamp in cast acrylic for bars, a crystal ball

mm | issue 2026-10-19 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15